

Optimizing e commerce systems

Monday, September 29, 2025 1:59 PM

Search Engine Optimization (SEO)

What is SEO?

- SEO means improving a website so that it **appears at the top of search results** when people search on Google or other search engines.
- The goal: **make it easy for users to find your website.**
- Marketers want their website link to appear in the **top 10 results**, because most people don't scroll beyond the first page.

Example:

Imagine you own a bakery in Kathmandu. If someone types "**best bakery in Kathmandu**" on Google, you want your bakery's website to appear on the first page. SEO helps make that happen.

How Search Engines Work

Search engines (like Google) have 3 main parts:

1) Spider (Crawler/Robot)

- A spider is a program that **visits websites automatically** to collect information.
- It looks at:
 - Website **URL**
 - **Page title**
 - **Keywords** (important words on the page)
 - **Links to other pages**
- These keywords are often put inside **meta tags** in HTML.

Example:

If your bakery's website has words like "cakes," "pastries," "Kathmandu bakery," the spider reads them and saves that information.

2) Index (Database)

- This is like the **library** of the search engine.
- The spider sends all the collected information here.
- The index keeps updating itself with new or changed web pages.
- It allows very **fast searching** across billions of web pages.

Example:

When Google's spider scans your bakery page, it adds it to Google's index under "cakes in Kathmandu." Later, if someone searches for cakes, Google can quickly find your website.

3) Search Utility

- This is the part users interact with (Google's search box).
- When someone types a search term, the search utility looks inside the index to find matching results.
- Then, it shows a list of relevant websites (called the **Search Engine Results Page – SERP**).

Example:

When someone types "birthday cake Kathmandu," Google checks its index and shows your bakery website if it's optimized well.

Types of SEO

1. On-Page SEO (Inside the website)

Focuses on what is **on your website**.

Techniques:

- Use **good keywords** (e.g., “birthday cakes in Kathmandu”).
- Write **quality content** (blogs, product descriptions, etc.).
- Use **titles and headings** properly.
- Add **internal links** (linking one page of your site to another).
- Optimize **URLs** (e.g., use www.bakerynepal.com/birthday-cakes instead of random numbers).
- Ensure the website is **mobile-friendly** and loads fast.

Example:

Your bakery website has a page called “Chocolate Birthday Cakes in Kathmandu” with good pictures, descriptions, and keywords. That’s strong on-page SEO.

2. Off-Page SEO (Outside the website)

Focuses on how **popular and trusted your website is** on the internet.

Techniques:

- **Link building** (getting other websites to link to your site).
- **Social media engagement** (sharing your site on Facebook, Instagram).
- **Influencer outreach** (a food blogger recommending your cakes).
- **Article submissions, forum postings, business profiles.**
- Google Business Profile (important for local searches).

Example:

- A famous Nepali food blogger writes an article: “*Top 5 Bakeries in Kathmandu*” and adds a link to your bakery.
- This backlink increases your credibility, and your ranking improves.

Summary Table: On-Page vs Off-Page SEO

On-Page SEO (What your site is about)	Off-Page SEO (How popular your site is)
Keywords, content quality, titles	Backlinks from other sites
Internal links, page speed	Social media engagement
Mobile friendly, navigation	Mentions by influencers, forums
Example: Optimizing your bakery website	Example: A blogger linking to your bakery

Page Rank (PR)

Meaning:

- **Page Rank (PR)** is an algorithm developed by Google to **rank websites in search results**.
- Named after **Larry Page** (Google’s co-founder).
- It measures the **importance of a webpage** based on how many other sites link to it and how trustworthy those links are.
- Assumption: **Important pages get more links.**
- Today, Google uses many algorithms, but PR was the **first and most famous one**.

Example:

Suppose Daraz’s website gets many links from trusted sites like Kantipur Online (ekantipur.com) and Tech blogs in Nepal.

- Google thinks: “If trusted sites are linking to Daraz, it must be important.”
- So, Daraz’s site appears higher in search results compared to a new e-commerce site with no backlinks.

SEO Activities to Improve Website Rank

1. Use Google Analytics to Track Metrics

- Google Analytics shows how your website is performing (visitors, location, devices, time spent, bounce rate).

- Marketers can analyze **which pages are attracting visitors** and **which need improvement**.

Example: Daraz finds that its “Mobile Phones” page gets high traffic, but “Clothes” has very low visitors. They use this info to improve the clothes section.

2. Optimize Existing Pages

- Review older content and improve it (better titles, keywords, images).
- Remove broken links or errors.

Example: Daraz updates its “Samsung Phones Price in Nepal” page to include 2025 models instead of showing outdated 2022 models.

3. Create Content Based on Keywords

- Use **relevant keywords** people actually search for.
- Avoid keyword stuffing (repeating keywords unnaturally).
- Build **useful content** around those keywords.

Example: Instead of repeating “cheap phone” 20 times, Daraz creates a blog post: “*Top 10 Budget Smartphones in Nepal (2025)*”.

4. Diversify Backlink Portfolio

- Get links from many different **relevant and trusted websites** (blogs, news sites, forums).
- Shows Google that your site is **authoritative and natural**.

Example: Daraz gets backlinks from Nepali Tech YouTubers, gadget blogs, and local newspapers → builds trust.

5. Optimize Images

- Use **alt text** (short description of image).
- Compress images (faster loading).
- Make sure images look good on mobile.

Example: An image of iPhone 15 has alt text = “*iPhone 15 Price in Nepal – Daraz*”. This helps Google show the image in searches.

6. Make Sure Site is Mobile-Friendly

- Most searches are from smartphones.
- Google prioritizes mobile-optimized websites.

Example: If Daraz’s site loads properly on desktop but is hard to use on mobile → it will rank lower.

7. Decrease Load Times

- A slow site = bad user experience.
- Use tools like **Google PageSpeed Insights** to fix speed issues.

Example: If Daraz takes 10 seconds to load product images, customers leave. After compressing images and using faster hosting, load time drops to 2 seconds.

8. Track Results

- Regularly check SEO progress (ranking, traffic, bounce rate).
- Adjust strategies accordingly.

Example: Daraz tracks keyword “Buy iPhone in Nepal.” If ranking improves from 10th to 3rd on Google, they know their SEO is working.

Recommendation Systems

Meaning:

- A **Recommendation System** is a tool that uses algorithms, data analysis, and AI to **predict a user’s interests** and suggest products/content.
- Goal: Make it easier for users to find items they like and help businesses sell more.

- Works by analyzing **user behavior and item features**.

Example:

If you buy a **T-shirt** on Daraz, the next time you log in, the site may recommend:

- More **T-shirts** (different colors/brands)
- Related items like **jeans, jackets, or shoes**

Data Used in Recommendation Systems:

1. **User ratings** (e.g., rating a Netflix movie 5 stars).
2. **Search queries** (e.g., searching “budget phone under 20,000 in Nepal”).
3. **Purchase history** (e.g., frequently buying sports shoes → site suggests more sportswear).

Advantages:

- **Increase Sales:** Customers buy more because they see relevant items.
 - Example: Buying a laptop → system recommends laptop bags, mouse, antivirus.
- **Faster Search:** Saves users from searching through thousands of products.
- **Customer Retention:** Sending personalized emails with recommended items keeps customers engaged.
- **Cross-Selling:** Customers often buy extra items.
 - Example: Amazon recommends “Frequently Bought Together” bundles.

Types of Recommendation Systems

1. Content-Based Systems

- Recommend items **similar to those the user liked before**.
- Uses **features of items** (color, brand, category, genre, etc.) and **user’s past choices**.
- Builds a **profile of the user** to suggest similar items.

Example:

- If a user buys a **Samsung Galaxy Phone**, the system recommends:
 - Other **Samsung phones**
 - Phones with similar features (same price range, RAM, or battery life)

2. Collaborative Filtering Systems

- Recommend items based on **what other similar users liked**.
- Assumption: *“If two users have liked the same item before, they might like similar items in the future.”*
- Common in **Netflix, YouTube, Amazon**.

Example:

- User A and User B both bought **T-shirts**.
- User A later buys **jeans**.
- System recommends **jeans** to User B as well.

Use of Recommendation Systems in E-commerce

- **Problem:** Huge number of products online makes it difficult for users to choose.
- **Solution:** Recommendation systems **filter and highlight the most relevant products** for each user.

Benefits in E-commerce:

1. **Decision-making support** → Helps customers choose quickly.
2. **Increased sales** → More personalized offers → More purchases.
3. **Customer loyalty** → People return because the site “understands” them.
4. **Better experience** → Users find items they might not even think of searching.

Example (Daraz Nepal):

- A customer buys a **budget phone**.
- Recommendation system suggests:
 - **Phone cover**

- **Power bank**
 - **Earbuds**
 - Other **budget-friendly phones**
- This makes shopping faster for the customer and increases Daraz's sales.